15. In two-way classification with 'm' treatments and 'n' blocks the degrees of freedom due to error is

(a) mn-1 (b) m-1 (c) n-1 (d) (m-1)(n-1)

16. If the calculated value of F is greater than the critical value at the given level of significance then the H_0 is

(a) Rejected (b) Not rejected

(c) Always true (d) Sometimes true

17. _____ and _____ causes are present in Analysis of Variance techniques

(a) Chance, error (b) Fixed, block

(c) Assignable, chance (d) Assignable, fixed

18. In ANOVA, the sample observations are

(a) dependent (b) independent

(c) equal (d) unequal

19. The correction factor is _____ in ANOVA (with the usual notations).

(a) $\frac{\sum T_{ij}^2}{n}$ (b) $\frac{\sum T_{i.}^2}{n}$

(c) $\frac{G^2}{n}$ (d) $\frac{\sum T_{i.}}{n}$

20. Mean Sum of Squares is the ratio of Sum of Squarea to its

(a) number of blocks (b) number of treatments

(c) degrees of freedom (d) total sum of squares

II. Give very short answers to the following questions.

21. What is Analysis of Variance?
22. Write the applications of F-statistic.
23. What are the assumptions of ANOVA?
24. Define: Between group variance and within group variance.
25. State the hypotheses used in one-way ANOVA.
26. What are the components in a two-way ANOVA?
27. Name the causes of variation?

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